

ZeroDrift · Slide review before submission

Every slide checked word by word against the code, the canonical numbers table (docs/PROJECT_STATE.md) and the SIH guidelines. Red / orange / blue numbers on each slide match the change list under it.

MUST fix

SHOULD fix

NICE to have

Score as an SIH judge (out of 10)

Slide	Score now	After fixes	One-line reason
Slide 1 · Title page	8	9	Clean and complete; only polish left.
Slide 3 · Technical Approach	7	9	Best content; typos, tiny text, generated pictures, missing "why chosen".
Slide 4 · Feasibility and Viability	6.5	8.5	Right story; old test count, unsourced 2.8 s, generated art.
Slide 5 · Impact and Benefits	6.5	8.5	Best design; unprovable claims, DRDO logo, no licence file.
Slide 6 · Research and References	6.5	8.5	Right content; old numbers, dead [link]s, "wins" wording.
Slide 2	not rated (as requested)		

Fix these first (all slides)

- Test count:** slides 4 and 6 say 73. Use **250 / 251**.
- "0 wrong locks":** always write **0 / 64 runs**.
- Generated pictures:** slide 3 (5 pictures, I made replacements), slide 4 (risk cards + centre), slide 5 (icons, DRDO seal).
- Unsourced numbers:** slide 4 "2.8 s under stress" → replace with 2.77 s p95. Slide 6 "six deliverables", "5,000 lines", "6 scenarios".
- Spelling:** REAQUIRE, detentions, instabilit, forthe, Github, Pyinstaller, TLES.
- Links:** the seven [link] buttons on slide 6 must open real pages.

Guidelines check

The SIH guidelines PDF has no rule on AI-generated images and no negative marking. Judging criteria listed there: novelty, complexity, clarity and detail in the prescribed format, feasibility, practicability, sustainability, scale of impact, user experience, future potential. Deadline 30 Sept 2026, no requests after it. Slide 3 should also say why each model or framework was chosen (change 15).

Slide 1 · Title page

8 / 10 → 9 / 10 after fixes

Page 1 - Done

SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID - 26169

Problem Statement Title - AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile FSOC Terminals

Theme - Smart Automation

PS Category - Software

Team ID - 148157

Team Name - ZeroDrift

1

ZERO-DRIFT

PRECISION TRACKING | SIH 2026 |

2

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SMART INDIA HACKATHON 2026

SIH

1

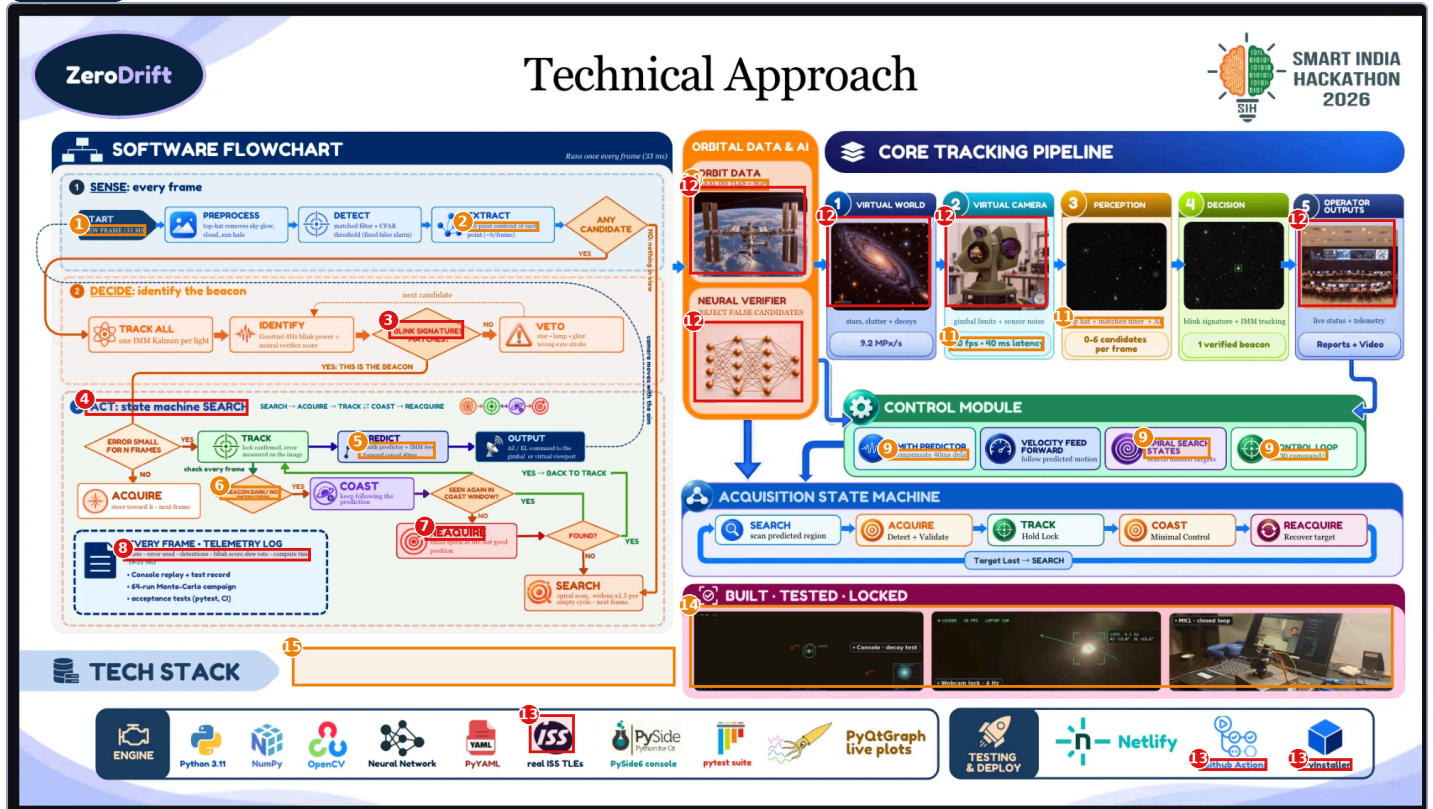
Judge: Clean, all required fields present (PS ID, title, theme, category, team ID, team name), both logos visible. Nothing wrong, only polish.

#	Where	Now → Change to	Why
1	NICE	PRECISION TRACKING SIH 2026 (stray " " at the end) → PRECISION TRACKING SIH 2026	
2	NICE	(nothing) → Jabalpur Engineering College · Team ZeroDrift · The pointing brain	

Slide 3 · Technical Approach

7 / 10

→ 9 / 10 after fixes



Judge: Strongest slide: a real flowchart, real proof strip, honest tech stack. Held back by leftover typos, unreadably small text in the ACT lane, and pictures that are AI-generated and are not our system (galaxy, military gimbal, control room, ISS, neural net). The guidelines also want the slide to say WHY each model/framework was chosen: add that line.

#	Where	Now → Change to	Why
1	SHOULD SENSE lane, START box	NEW FRAME (33 MS) → NEW FRAME (33 ms)	Unit case.
2	SHOULD EXTRACT box	sub pixel centroid of each point (~6/frame) → sub-pixel centroid of each point (~6 / frame)	Hyphen.
3	MUST DECIDE lane, diamond	BLINK SIGNATURES MATCHES? → BLINK SIGNATURE MATCHES?	Grammar (plural + singular verb).
4	MUST ACT lane heading	ACT: state machine SEARCH → ACT: state machine	SEARCH is repeated: the path next to it already lists all five states.
5	SHOULD PREDICT box	smith predictor + IMM feed forward cancel 40ms → Smith predictor + IMM feed-forward, cancels 40 ms	Names take a capital; unit needs a space.
6	SHOULD ACT lane, small diamond	BEACON DARK/ NOT DETECTED? → BEACON DARK / NOT DETECTED?	Spacing.
7	MUST ACT lane, red box	REQUIRE → REACQUIRE	Spelling (still wrong here; fixed in the bottom bar).
8	MUST EVERY FRAME · TELEMETRY LOG	state - error urad - detentions - blink score slew rate - compute time → state · error (μrad) · detections · blink score · slew rate · compute time	"detentions" is a wrong word, "urad" should be μrad, and one separator is missing.
9	SHOULD CONTROL MODULE (three cards)	Compensate 40ms delay SPIRAL SEARCH STATES ~ 30 command/s → Compensates 40 ms delay SPIRAL SEARCH ~30 commands/s	Grammar, unit spacing, plural.
10	SHOULD ORBIT DATA sub-title	REAL ISS TLES + SGP4 → Real ISS TLEs + SGP4	TLE plural is TLEs.

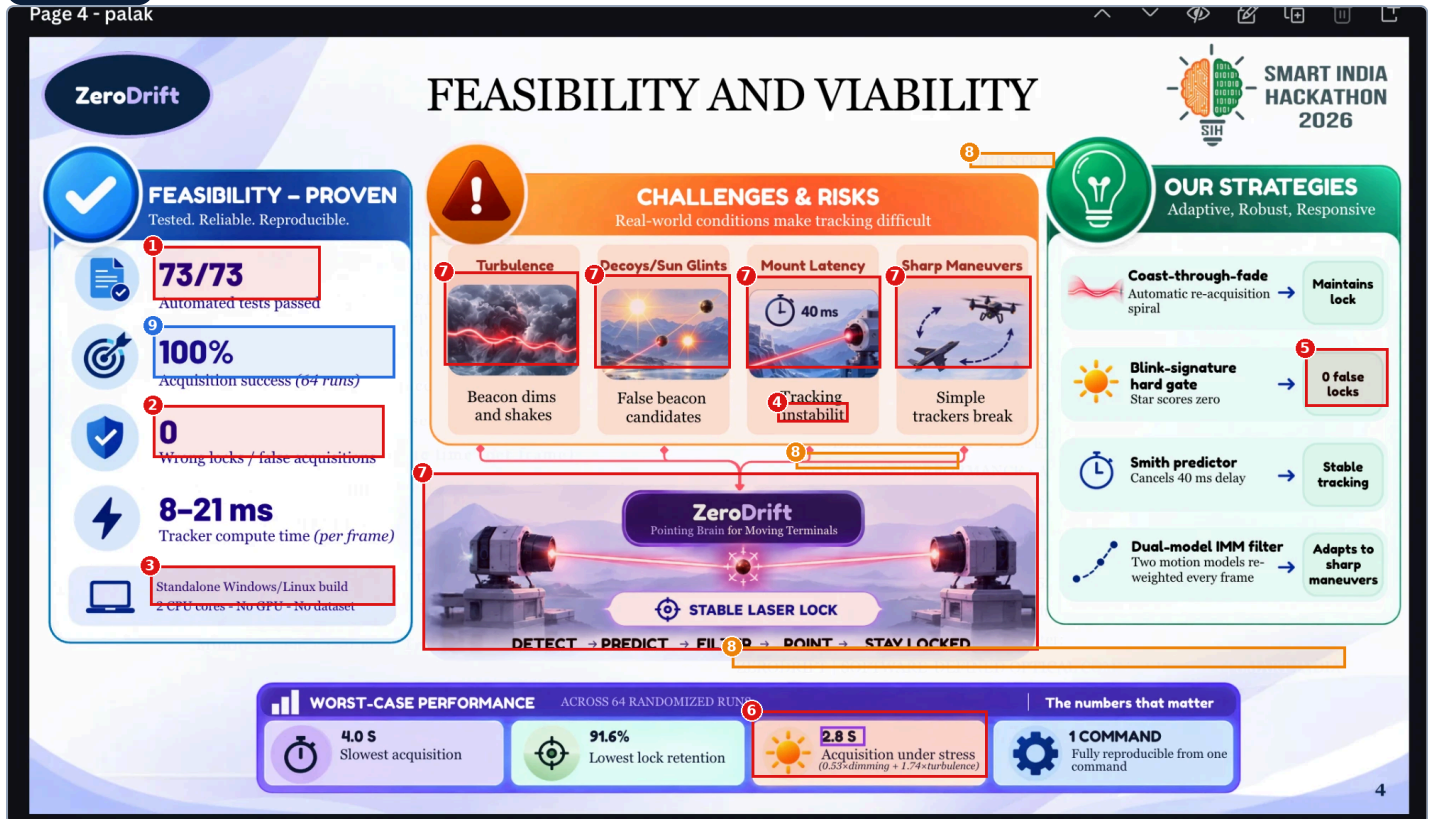
11	SHOULD Pipeline cards 2 and 3	30 fps · 40ms latency top hat + matched filter + AI → 30 fps · 40 ms latency top-hat + matched filter + AI	Unit spacing, hyphen.
12	MUST Five pictures: Orbit data, Neural verifier, Virtual world, Virtual camera, Operator outputs	AI-generated stock images (ISS photo, neural-net art, galaxy, military gimbal, control room) → Use my 5 images (docs/submission/slide3_own_images/): 1_orbit_data → Orbit Data, 2_neural_verifier → Neural Verifier, 3_virtual_world → Virtual World, 4_virtual_camera → Virtual Camera, 5_operator_outputs → Operator Outputs	They look like real photos but are generated, and none is our system. The replacements come from our own engine and Console. Perception and Decision pictures are already engine renders: keep.
13	MUST TECH STACK bar	round "ISS" logo · Github Actions · Pyinstaller → Replace the round logo with a plain text badge "ISS TLEs"; write GitHub Actions and PyInstaller	The round logo is a company logo (ISS A/S), not the space station. Product names have fixed spelling.
14	SHOULD BUILT · TESTED · LOCKED strip	first picture very dark and tiny → Use the 3 proof photos (2000×694): 1_console_decoy_test, 2_webcam_lock_4hz, 3_mk1_closed_loop	Readable at phone size, small labels in empty corners.
15	SHOULD Free space right of the TECH STACK heading	(empty) → Chosen for: CPU-only speed (8–21 ms per 33 ms frame) · blink-rate check gives 0 / 64 wrong locks · neural verifier AUC 0.957 vs 0.900 classical · one command reproduces every figure	Guidelines: the technical slide must state why each model/framework was chosen and how it measurably helps accuracy and scalability. Every number is in the canonical table.

Slide 4 · Feasibility and Viability

6.5 / 10

→ 8.5 / 10 after fixes

Page 4 - palak



Judge: Good structure (proof, risks, strategies, worst case) and the right story. But it repeats the old test count, uses "0 false locks" without saying "of 64 runs", has one number with no source (2.8 s), a wrong platform claim (Linux), a typo, ghost background text, and generated pictures in the risk cards.

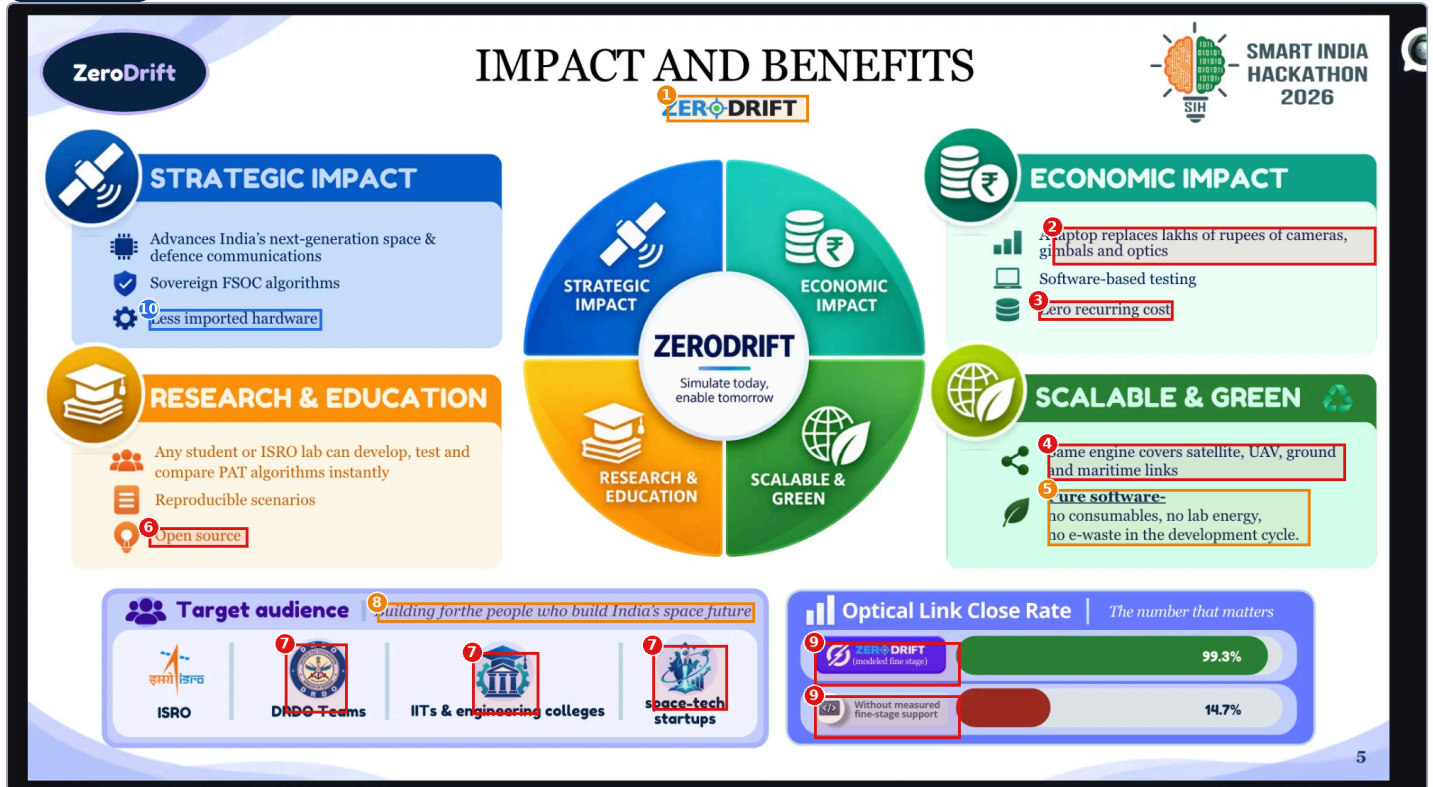
#	Where	Now → Change to	Why
1	MUST FEASIBILITY – PROVEN, first row	73/73 Automated tests passed → 250 / 251 Automated tests passing (1 skipped)	Verified count on 28 Sept. Must match slide 6.
2	MUST FEASIBILITY – PROVEN, third row	0 Wrong locks / false acquisitions → 0 / 64 runs with a wrong-light lock	0 is true for the 64-run campaign, not for every frame ever (the decoy demo run had 1 frame on a decoy).
3	MUST Bottom of the left box	Standalone Windows/Linux build · 2 CPU cores · No GPU · No dataset → Standalone Windows/macOS build · 2 CPU cores · No GPU · No external dataset	The build workflow makes Windows and macOS apps, not Linux. The neural verifier is trained on our own simulator, so say "no external dataset".
4	MUST CHALLENGES & RISKS, Mount Latency card	Tracking instabilit → Tracking instability	Missing letter.
5	MUST OUR STRATEGIES, second row	0 false locks → 0 / 64 wrong locks	Same reason as change 2.
6	MUST WORST-CASE PERFORMANCE, third box	2.8 S Acquisition under stress (0.53×dimming + 1.74×turbulence) → 2.77 s 95th-percentile acquisition (64 runs)	No source for 2.8 s / 0.53× / 1.74× exists in the repo or canonical table. A judge who asks will not get an answer. 2.77 s is canonical.
7	MUST The four risk pictures (storm, sun glints, mount, jets) and the big centre picture with two telescopes	AI-generated stock art → Replace with flat icons, or with our own plots: turbulence trace, decoy diamonds (Console), latency graph, IMM plot. Centre: use the Console screenshot with the LOCKED ring.	Generated images that look real hurt credibility; the centre picture is not our hardware.
8	SHOULD Faint text behind the panels	"OUR STRA...", "ZERODRIFT SOFTWARE-DEFINED OPTICAL COM...", "WORST-CASE PERFORMANCE" → Delete these ghost layers (leftover from the background image)	Looks like an export mistake.

9	NICE	100% Acquisition success (64 runs) → 100% Acquisition success (64 runs) · median 1.27 s	
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Slide 5 · Impact and Benefits

6.5 / 10

→ 8.5 / 10 after fixes



Judge: Visually the best-looking slide and the right four impact areas. The 99.3% vs 14.7% bars are correct (canonical). Loses marks for claims we cannot prove (ground and maritime, "less imported hardware"), a duplicated logo, the DRDO logo, generated icons, a missing space, and "Open source" with no licence file in the repo.

#	Where	Now → Change to	Why
1	SHOULD Under the title	second ZERODRIFT logo → Delete it	The logo is already top-left. Duplicate looks like a leftover.
2	MUST ECONOMIC IMPACT, first bullet	A laptop replaces lakhs of rupees of cameras, gimbals and optics → A laptop replaces a ₹15–50 lakh optical test bench	Number from docs/economic_feasibility.md.
3	MUST ECONOMIC IMPACT, third bullet	Zero recurring cost → Near-zero running cost (₹0.22 for a full 64-run campaign)	Zero is an absolute claim; ₹0.22 electricity is the measured figure.
4	MUST SCALABLE & GREEN, first bullet	Same engine covers satellite, UAV, ground and maritime links → Same engine runs satellite (LEO) and UAV-relay scenarios; ground and maritime are next	We have scenario files for LEO passes and a UAV relay only. Do not claim what is untested.
5	SHOULD SCALABLE & GREEN, second bullet	Pure software- no consumables, no lab energy, no e-waste in the development cycle. → Pure software: no consumables, no lab energy, no e-waste.	Dangling hyphen and underline look like a broken edit.
6	MUST RESEARCH & EDUCATION, third bullet	Open source → Open source (MIT licence) → I can add the LICENSE file to the repo	The repository has no LICENSE file, so "open source" is not yet legally true.
7	MUST Target audience strip	DRDO seal + generated icons for IITs and space-tech startups → Keep the real ISRO logo; write DRDO, IITs and space-tech startups as text with simple flat icons	Using an organisation seal implies endorsement. The two icons are generated art.
8	SHOULD Target audience heading	Building for the people who build India's space future → Building for the people who build India's space future	Missing space.

9	MUST Optical Link Close Rate labels	ZeroDrift (modeled fine stage) / Without measured fine-stage support → ZeroDrift + modelled fine stage (99.3%) / Coarse stage alone (14.7%) + small line: "Link- budget model, technical report"	99.3% and 14.7% are canonical, but the second label is confusing. "Modelled" must stay visible.
10	NICE	Less imported hardware → Reduces dependence on imported test hardware	

Slide 6 · Research and References

6.5 / 10

→ 8.5 / 10 after fixes

RESEARCH AND REFERENCES

Research foundations

Kaymak et al.,
"A Survey on Acquisition, Tracking and Pointing Mechanisms for Mobile FSO,"
— IEEE Comm. Surveys & Tutorials, 2018

Kaushal & Kaddoum,
"Optical Communication in Space: Challenges and Mitigation Techniques,"
— IEEE Comm. Surveys & Tutorials, 2017

Blackman & Popoli,
Design and Analysis of Modern Tracking Systems
— IMM filtering, CFAR detection, track management

Bar-Shalom et al.,
Estimation with Applications to Tracking and Navigation — Kalman/TMM theory

Smith, O.J.M.,
"Closer Control of Loops with Dead Time" — The Smith predictor, 1957.

Vallado & Crawford,
"SGP4 Orbit Determination" — AIAA 2008; real ISS TLE Propagation

Andrews & Phillips,
Laser Beam Propagation through Random Media — Turbulence & Scintillation models

Our work – Open and Verifiable

Repository github.com/pranshukesavmishra/SIH-2026
Source code • Data and Resources • Open for Everyone

Live demo zerodrift-fsoc-pat.netlify.app
Try it live • explore scenarios and results

Inside [all source \(5,000+ lines\) • 73 automated tests • 6 validated scenarios • Monte-Carlo campaign logs • technical report • user manual • demo videos](#)

Reproducibility Every figure regenerates from one command; every random draw is logged and replayable.

Why ZeroDrift wins on this problem

- ✓ All six mandatory deliverables already exist today
- ✓ Judged numbers, honestly measured against hidden ground truth
- ✓ AI where it earns its place — benchmarked against classical methods

Compared with existing approaches

Feature	ZERO-DRIFT	MATLAB sim	Traditional
Zero hardware cost No special equipment required	✓	✓	✗
Real-time closed loop & Live console See it run. Live.	✓	✗	✓
Decoy & Blink-Signature Distinction Robust against spoofing	✓	✗	✗
Open Source & Reproducible Run with a single command	✓	✗	✗

Judge: The right content for an "open and verifiable" claim: papers, repo, live demo, comparison. Held back by old numbers (73 tests, 6 scenarios, 5,000 lines), links that are only the word [link], a "wins" headline, unexplained ticks in the comparison table, and publisher logos used as decoration.

#	Where	Now → Change to	Why
1	MUST Our work, "Inside" row	Full source (5,000+ lines) • 73 automated tests • 6 validated scenarios → Full source (8,500+ lines of Python) • 250 automated tests • 8 scenario files	Counted from the repo: 8,580 lines in src/, 250 tests passing, 8 scenario files.
2	MUST Research foundations, right column	[link] ×7 (only the word) → Make each one a real clickable URL: search the paper title on IEEE Xplore / publisher site and paste its DOI link; in PowerPoint: select text → Ctrl+K	A reference that cannot be opened counts as no reference. I did not invent DOIs.
3	SHOULD Blackman & Popoli / Bar-Shalom / Andrews & Phillips / Smith	no year, no publisher → Blackman & Popoli, 1999, Artech House · Bar-Shalom et al., 2001, Wiley · Andrews & Phillips, Laser Beam Propagation through Random Media, SPIE Press · Smith, 1957, Chem. Eng. Progress (check against each book's title page)	Consistent citations look serious.
4	SHOULD Bottom-left logo block	IEEE Xplore, "1957 Smith Predictor", Wiley, artech house, SPIE, AIAA logos → Delete the block; use the space to make the reference text bigger	Publisher logos are decoration and imply affiliation; text is not readable at phone size.
5	MUST Dark box title	Why ZeroDrift wins on this problem → Why ZeroDrift fits this problem	"Wins" sounds overconfident to judges.
6	MUST Dark box, first tick	All six mandatory deliverables already exist today → Every mandatory deliverable of the problem statement is implemented	Our own docs say five mandatory deliverables, the slide says six. Do not give a number unless it matches the PS text.
7	SHOULD Comparison table	Compared with existing approaches (ticks and crosses, MATLAB sim ticked "Zero hardware cost") → Title: "Our assessment vs typical approaches"; MATLAB row 1 → "No special hardware"; add footnote: "Based on our review, not a head-to-head benchmark."	MATLAB is not free; a one-sided table with no footnote invites a challenge.